

Vectors in $\mathbb{R}^3$

1. Vectors and their vector structure

Def. A vector $\vec{v}$ in space is characterized by three data:

- magnitude / length ($\|\vec{v}\|$)
- direction
- sense

Obs. Vectors are added by the triangle law or the parallelogram law.

Addition postulates:

- 1) $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$
- 2) $\vec{u} + \vec{v} = \vec{v} + \vec{u}$
- 3) $\exists \vec{0}$ s.t. $\vec{0} + \vec{v} = \vec{v}$
- 4) $\exists (-\vec{v})$ s.t. $\vec{v} + (-\vec{v}) = \vec{0}$

Denote : $\mathcal{V}_3$ the set of all vectors in space.

Obs. Then $(\mathcal{V}_3, +)$ is a commutative group.

Scalar multiplications:

- 1) $1 \cdot \vec{a} = \vec{a}$
- 2) $n \cdot (t \cdot \vec{a}) = (nt) \cdot \vec{a}$
- 3) $(n+t) \cdot \vec{a} = n \cdot \vec{a} + t \cdot \vec{a}$
- 4) $n \cdot (\vec{a} + \vec{b}) = n \cdot \vec{a} + n \cdot \vec{b}$

Obs. Numbers n, t are called scalars. $t \cdot \vec{v}$ is a

scalar multiple of $\vec{v}$.

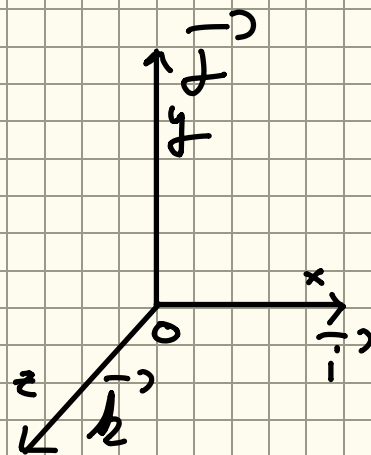
Def. Fix mutually perpendicular axes Ox, Oy, Oz forming a right-handed frame, with unit vectors $\vec{i}, \vec{j}, \vec{k}$. Every vector $\vec{v}$ is written as:

$$\vec{v} = a\vec{i} + b\vec{j} + c\vec{k}, \quad a, b, c \in \mathbb{R}$$

Properties:

$$1) \|a \cdot \vec{v}\| = |a| \cdot \|\vec{v}\|$$

$$2) \|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$$



2. Inner product

Def. The scalar (inner) product is the number:
 $\vec{a} \cdot \vec{b} = \|\vec{a}\| \cdot \|\vec{b}\| \cdot \cos \theta, \quad \theta = (\vec{a}, \vec{b})$

Obs. With $\vec{a} \cdot \vec{b} = 0$ whenever $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$

Properties:

$$1) \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

$$2) \vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

$$3) (n \cdot \vec{a}) \cdot \vec{b} = n(\vec{a} \cdot \vec{b})$$

$$4) \vec{a} \cdot \vec{a} \geq 0 \Leftrightarrow \vec{a} = \vec{0}$$

Obs. Since $\vec{a} \cdot \vec{a} = \|\vec{a}\|^2$:

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \cdot \|\vec{b}\|},$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow a \perp b$$

$$\vec{i}^2 = \vec{j}^2 = \vec{k}^2 = 1$$

$$\vec{i} \cdot \vec{j} = \vec{i} \cdot \vec{k} = \vec{j} \cdot \vec{k} = 0$$

{ The basis is orthonormal

Ob. Let $\vec{a} = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}$ and $\vec{b} = b_1 \vec{i} + b_2 \vec{j} + b_3 \vec{k}$:

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\|\vec{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

$$\cos \theta = \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \cdot \sqrt{b_1^2 + b_2^2 + b_3^2}}$$

$$a \perp b \Leftrightarrow a_1 b_1 + a_2 b_2 + a_3 b_3 = 0$$

3. Vector (cross) product

Def. The vector product $\vec{a} \times \vec{b}$ is the vector characterized by:

$$1) \|\vec{a} \times \vec{b}\| = \|\vec{a}\| \cdot \|\vec{b}\| \cdot \sin \theta$$

$$2) (\vec{a} \times \vec{b}) \perp \vec{a} \text{ and } (\vec{a} \times \vec{b}) \perp \vec{b}$$

$$3) \{\vec{a}, \vec{b}, \vec{a} \times \vec{b}\} \text{ is oriented as } \{\vec{i}, \vec{j}, \vec{k}\}$$

(right-hand rule)

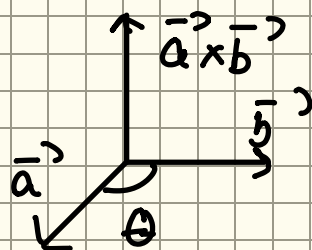
Properties:

$$1) \vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$$

$$2) (n \cdot \vec{a}) \times \vec{b} = \vec{a} \times (n \cdot \vec{b}) = n(\vec{a} \times \vec{b})$$

$$3) \vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$$

$$4) \vec{a} \times \vec{0} = \vec{0}$$



$$5) \vec{a} \times \vec{a} = \vec{0}$$

$$6) \vec{a} \times \vec{b} = \vec{0} \Leftrightarrow \vec{a} \parallel \vec{b}$$

$$7) \|\vec{a} \times \vec{b}\| = \text{area of the parallelogram of } \vec{a} \text{ and } \vec{b}.$$

Multiplication table for $\vec{i}, \vec{j}, \vec{k}$:

$\times$	$\vec{i}$	$\vec{j}$	$\vec{k}$
$\vec{i}$	1	$\vec{k}$	$-\vec{j}$
$\vec{j}$	$-\vec{k}$	1	$\vec{i}$
$\vec{k}$	$\vec{j}$	$-\vec{i}$	1

$$\vec{a} \times \vec{b} = (a_2 b_3 - a_3 b_2) \vec{i} + (a_3 b_1 - a_1 b_3) \vec{j} + (a_1 b_2 - a_2 b_1) \vec{k}$$

Determinant formula:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

4. Triple product identity:

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$$

Obv. This has no geometrical meaning. The product always lies in the plane spanned by $\vec{b}$ and $\vec{c}$ and is consistent with the right hand side.

5. Triple scalar product

Def. The triple scalar product of $\vec{a}, \vec{b}, \vec{c}$ is:

$$(\vec{a}, \vec{b}, \vec{c}) = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Properties:

$$1) (\vec{a}, \vec{b}, \vec{c}) = (\vec{c}, \vec{a}, \vec{b}) = (\vec{b}, \vec{c}, \vec{a})$$

$$2) (\vec{a}, \vec{b}, \vec{c}) = -(\vec{a}, \vec{c}, \vec{b})$$

$$3) (0 \cdot \vec{a}, \vec{b}, \vec{c}) = 0 (\vec{a}, \vec{b}, \vec{c})$$

$$4) (\vec{u} + \vec{v}, \vec{b}, \vec{c}) = (\vec{u}, \vec{b}, \vec{c}) + (\vec{v}, \vec{b}, \vec{c})$$

Ob. 1) $|(\vec{a}, \vec{b}, \vec{c})|$ is the volume of the parallelepiped spanned by $\vec{a}, \vec{b}, \vec{c}$.

$$2) (\vec{a}, \vec{b}, \vec{c}) = 0 \Leftrightarrow \vec{a}, \vec{b}, \vec{c} \text{ coplanar}$$

$$3) (\vec{a}, \vec{b}, \vec{c}) \text{ has positive orientation if}$$

$$(\vec{a}, \vec{b}, \vec{c}) > 0$$

$$4) (\vec{a}, \vec{b}, \vec{c}) \text{ has negative orientation if}$$

$$(\vec{a}, \vec{b}, \vec{c}) < 0.$$

